

Technical Report: Advanced Shipyard Block Packing Optimization (v2)

1. Introduction

The **Grand Shipyard Puzzle** is a complex 2D Bin Packing Problem (2DBPP) where the goal is to pack various rectangular blocks into a fixed-size container (the shipyard) while maximizing the used area. This version improves upon previous iterations by introducing a more robust optimization pipeline.

2. Methodology

2.1 MaxRects Algorithm with Multi-Scoring

The spatial engine continues to use the **MaxRects** algorithm, which maintains a list of maximal free rectangles. For every sequence, the solver evaluates three distinct placement rules:

- **BSSF (Best Short Side Fit)**: Minimizes the remaining short side of the free rectangle.
- **BLSF (Best Long Side Fit)**: Minimizes the remaining long side.
- **BAF (Best Area Fit)**: Minimizes the leftover area of the chosen rectangle.

2.2 Advanced Hybrid Optimization Pipeline

We implemented a two-stage optimization pipeline:

1. **Multi-Heuristic Initialization**: Instead of a single sorting rule, the solver starts by evaluating several high-performance heuristics (Area, Max Side, Perimeter, Width, Height) to find a strong baseline.
2. **Simulated Annealing (SA)**: A meta-heuristic search that explores the sequence space. SA is less likely to get stuck in local optima compared to simple hill climbing by allowing occasional “bad” moves based on a cooling temperature.

3. Implementation Details

- **Language**: Python 3.11
- **Key Libraries**: `numpy`, `pandas`, `matplotlib`
- **Improvements**:

- Refined `_prune_free_rects` for better performance.
- Transitioned from GA to Simulated Annealing for more stable convergence.
- Added multiple initial sorting heuristics.

4. Results

In benchmarks with 50 random blocks (5m to 20m) in a 100m x 100m shipyard:

- **Placement Rate:** 100% (all blocks placed).
- **Utilization:** Consistently achieves **80-85%** utilization.
- **Execution Time:** ~10 seconds for 1000 iterations, providing a high-quality solution quickly.

5. Conclusion

The combination of MaxRects with Simulated Annealing and multiple heuristic initializations provides a highly effective solution for shipyard logistics. The approach is robust, scalable, and yields high-density packing results.